

$$\hat{f}(\xi) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-i\xi x} dx$$

$$\widehat{f(x) e^{itx}}(\xi) = \hat{f}\left(\xi - t\right)$$

$$\widehat{f(ax)}(\xi) = \frac{1}{a} \hat{f}\left(\frac{\xi}{a}\right)$$

$$\widehat{f(x-t)}(\xi) = \hat{f}(\xi) e^{-it\xi}$$

$$\text{Inverz: } f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \hat{f}(\xi) e^{i\xi x} d\xi$$

$$\hat{\hat{f}}(x) = f(-x)$$

Tabela znanih FT:

$f(x)$	$\hat{f}(\xi)$
$e^{- x }$	$\sqrt{\frac{2}{\pi}} \frac{1}{1+\xi^2}$
$e^{-\frac{x^2}{2}}$	$e^{-\frac{\xi^2}{2}}$
$e^{-\frac{a^2 x^2}{2}}$	$\frac{1}{a} e^{-\frac{\xi^2}{2a^2}}$ $a > 0$
$\chi_{[-a,a]}(x)$	$\sqrt{\frac{2}{\pi}} \frac{\sin(a\xi)}{\xi}$

$$\textcircled{1} f(x) = \frac{e^{ix}}{x^2 + 2x + 5}$$

$$\hat{f}(\xi) = \widehat{\frac{1}{x^2 + 2x + 5}}(\xi - 1) = \widehat{\frac{1}{(x+1)^2 + 4}}(\xi - 1) = \frac{1}{x^2 + 4}(\xi - 1) e^{+i\xi} = \frac{1}{4} \frac{1}{(\frac{x}{2})^2 + 1}(\xi - 1) e^{i\xi} =$$

$$= \frac{1}{4} e^{i\xi} \cdot 2 \frac{1}{1+x^2} (2(\xi - 1)) =$$

$$= \boxed{\frac{1}{2} e^{i\xi} \sqrt{\frac{2}{\pi}} e^{-|2(\xi - 1)|}}$$

$$\left[\begin{aligned} \widehat{e^{-|x|}}(\xi) &= \sqrt{\frac{2}{\pi}} \frac{1}{1+\xi^2} / \cdot 1 \\ e^{-|x|} &= \sqrt{\frac{2}{\pi}} \frac{1}{1+\xi^2}(x) \end{aligned} \right.$$

$$\textcircled{2} f(x) = e^{-x^2} \cos(2x)$$

$$\hat{f}(\xi) = \dots = \frac{1}{2\sqrt{2}} e^{-\frac{(\xi-2)^2}{4}} + \frac{1}{2\sqrt{2}} e^{-\frac{(\xi+2)^2}{4}}$$

ODVOD FT

$$\widehat{f'(x)}(\xi) = i\xi \hat{f}(\xi)$$

$$\widehat{f''(x)}(\xi) = -\xi^2 \hat{f}(\xi)$$

KONVOLUCIJA

$$(f * g)(t) = \int_{-\infty}^{\infty} f(x-t)g(t)dt$$

$$\widehat{f * g}(\xi) = \sqrt{2\pi} \hat{f}(\xi) \hat{g}(\xi)$$

$$\hat{f}g(\xi) = \frac{1}{\sqrt{2\pi}} (\hat{f * g})(\xi)$$

• kako podobna sta grafa $g(t)$ in premaknjene $f(x-t)$ (+prezračene)

• $f * g$ ima maksimum tam, kamor moramo premakniti f , da bosta f in g najbolj podobni

③ Poišči f , ki reši enačbo

$$\int_{-\infty}^{\infty} f(t)f(x-t) dt = \frac{1}{1+x^2}$$

$$(f * f)(x) = \frac{1}{1+x^2} \quad / \wedge$$